Project1

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1. Introduction

LEGO sets are a global toy brand and a major collector's market, understanding what drives the price of a LEGO set has practical relevance for consumers, collectors, and manufacturers. The Brickset dataset offers a rich opportunity to explore how features related to price impact the market. By analyzing these variables and applying model selection techniques, the goal is to identify the most influential predictors of LEGO set price and construct a model that prioritizes not only predictive accuracy but also interpretability. This analysis will also allow us to examine larger patterns in LEGO set design and pricing across themes and release years.

2. Research Questions

The Brickset LEGO dataset provides detailed information about LEGO sets including price, piece count, themes, year released, and other characteristics. Because LEGO sets vary widely in complexity and price or market value, this dataset offers an opportunity to explore what factors drive the price of a set and whether these relationships can be modeled effectively.

Our primary research question is:

1. What factors most strongly influence the retails price of a LEGO set?

To support this, we also consider several smaller questions:

- Does piece count explain most of the variation in price?
- Does the year of release influence the price after controlling for all other variables?

To address these questions, multiple linear regression, best subsets selection, and stepwise regression to identify the most influential predictors and construct a final model balances simplicity and predictive accuracy.

3. Data Description

Before conducting any modeling, it is important to clearly outline the variables included in the analysis and the roles they play. This section provides an overview of the dataset's structure, identifies the response and predictor variables, and explains how each variable contributes to understanding what drives LEGO set retail prices

```
variable_table <- data.frame(
  Variable = c(
    "US_retailPrice", "pieces", "minifigs", "weight",
    "height", "width", "depth", "agerange_min",
    "rating", "reviewCount", "theme", "themeGroup", "category"
  ),

  Role = c(
    "Response",
    rep("Quantitative Predictor", 9),
    rep("Categorical Predictor", 3)
  ),

  Data_Type = c(
    "Numeric", "Numeric", "Numeric", "Numeric", "Numeric",
    "Numeric", "Numeric", "Numeric", "Numeric", "Numeric",
    "Categorical", "Categorical", "Categorical"
  ),

  Description = c(
    "Original U.S. retail price of the LEGO set in U.S. dollars",
    "Total number of LEGO pieces",
    "Number of minifigures included in the set",
    "Weight of the LEGO set",
    "Height of the LEGO set",
    "Width of the LEGO set",
    "Depth of the LEGO set",
    "Minimum recommended age for the set",
    "Average user rating for the set",
    "Number of user reviews",
    "Specific LEGO theme associated with the set",
    "Broader grouping of LEGO themes",
    "Category or type of LEGO set"
  )
)
variable_table
```

```
         Variable                    Role   Data_Type
1  US_retailPrice                Response     Numeric
2          pieces Quantitative Predictor     Numeric
3        minifigs Quantitative Predictor     Numeric
4          weight Quantitative Predictor     Numeric
5          height Quantitative Predictor     Numeric
6           width Quantitative Predictor     Numeric
7           depth Quantitative Predictor     Numeric
8    agerange_min Quantitative Predictor     Numeric
9          rating Quantitative Predictor     Numeric
10    reviewCount Quantitative Predictor     Numeric
11          theme  Categorical Predictor Categorical
12     themeGroup  Categorical Predictor Categorical
13       category  Categorical Predictor Categorical
                                                 Description
1  Original U.S. retail price of the LEGO set in U.S. dollars
2                                Total number of LEGO pieces
3                  Number of minifigures included in the set
4                                     Weight of the LEGO set
5                                     Height of the LEGO set
6                                      Width of the LEGO set
7                                      Depth of the LEGO set
8                          Minimum recommended age for the set
9                              Average user rating for the set
10                                    Number of user reviews
11               Specific LEGO theme associated with the set
12                           Broader grouping of LEGO themes
13                              Category or type of LEGO set
```

4. Data Cleaning

To prepare the dataset for analysis, we performed several data cleaning steps to ensure accuracy and consistency across all observations. This process involved identifying missing values and removing incomplete cases. Cleaning the data in this way allows us to work with a reliable subset of LEGO sets and prevents missing or inconsistent information from influencing the results of our exploratory analysis and regression models. The cleaned dataset, referred to as *data_complete*.

```r
#load data and read csv
df <- read.csv("2025.csv")

#create a table that counts all the missing data
variables <- c(
  "US_retailPrice",
  "pieces",
  "minifigs",
  "weight",
  "height",
  "width",
  "depth",
  "agerange_min",
  "rating",
  "reviewCount",
  "theme",
  "themeGroup",
  "category"
)

missing_table <- data.frame(
  Variable = variables,
  Missing = colSums(is.na(df[variables]))
)

missing_table
```

```
                    Variable Missing
US_retailPrice US_retailPrice     448
pieces               pieces     271
minifigs           minifigs     621
weight               weight     893
height               height     421
width                 width     421
depth                 depth     430
agerange_min   agerange_min     270
rating               rating       0
reviewCount     reviewCount       0
theme                 theme       0
themeGroup       themeGroup       1
category           category       0
```

```r
#create a new table that omits all the missing data
variables <- c(
  "US_retailPrice",
  "pieces",
  "minifigs",
  "weight",
  "height",
  "width",
  "depth",
  "agerange_min",
  "rating",
  "reviewCount",
  "theme",
  "themeGroup",
  "category"
)

data_complete <- df[complete.cases(df[variables]), ]
nrow(df)
```

```
[1] 1079
```

```r
nrow(data_complete)
```

```
[1] 61
```

```r
colSums(is.na(data_complete[variables]))
```

```
US_retailPrice         pieces       minifigs         weight         height
             0              0              0              0              0
         width          depth   agerange_min         rating    reviewCount
             0              0              0              0              0
         theme     themeGroup       category
             0              0              0
```

5. Exploratory Data Analysis

Summary statistics show substantial variation in LEGO set characteristics, especially price, piece count, and weight. Price and piece count are right-skewed, indicating that a small number

of large sets pull the mean upward. Ratings and review counts contain many zeros, suggesting limited user feedback for many sets. The descriptive statistics support further investigation into how physical characteristics relate to retail price.

```
# quantitative variables

quantitative_vars <- c(
  "US_retailPrice",
  "pieces",
  "minifigs",
  "weight",
  "height",
  "width",
  "depth",
  "agerange_min",
  "rating",
  "reviewCount"
)

# Select quantitative variables
correlation_vars <- data_complete[quantitative_vars]

# Calculate the correlation matrix
correlation_matrix <- cor(
  correlation_vars,
  use = "complete.obs"
)

correlation_matrix
```

```
               US_retailPrice    pieces  minifigs    weight    height     width
US_retailPrice      1.0000000 0.9106009 0.7786630 0.9671660 0.8191912 0.7984531
pieces              0.9106009 1.0000000 0.7555396 0.8658698 0.6260305 0.6528074
minifigs            0.7786630 0.7555396 1.0000000 0.7486760 0.6631737 0.6829421
weight              0.9671660 0.8658698 0.7486760 1.0000000 0.8464519 0.8272257
height              0.8191912 0.6260305 0.6631737 0.8464519 1.0000000 0.8323138
width               0.7984531 0.6528074 0.6829421 0.8272257 0.8323138 1.0000000
depth               0.7667257 0.6624670 0.5304681 0.8274525 0.7266921 0.7192446
agerange_min        0.5728106 0.7147580 0.3859962 0.4524908 0.3079079 0.3201698
rating              0.2523957 0.3476234 0.2676433 0.2636961 0.1064871 0.1294318
reviewCount         0.2599065 0.3492340 0.2701237 0.2828723 0.1505186 0.1648387
                    depth agerange_min    rating reviewCount
```

```
US_retailPrice 0.7667257    0.5728106 0.2523957   0.2599065
pieces         0.6624670    0.7147580 0.3476234   0.3492340
minifigs       0.5304681    0.3859962 0.2676433   0.2701237
weight         0.8274525    0.4524908 0.2636961   0.2828723
height         0.7266921    0.3079079 0.1064871   0.1505186
width          0.7192446    0.3201698 0.1294318   0.1648387
depth          1.0000000    0.4066463 0.1718117   0.2724942
agerange_min   0.4066463    1.0000000 0.3509209   0.3342968
rating         0.1718117    0.3509209 1.0000000   0.5944375
reviewCount    0.2724942    0.3342968 0.5944375   1.0000000
```

The correlation matrix shows strong positive relationships between LEGO set retail price and several measures of set size and complexity. The strongest correlation is between US retail price and weight (r = 0.967), followed by the number of pieces (r = 0.911). This suggests that heavier sets and sets containing more pieces tend to have higher retail prices. Price also has strong positive correlations with height (r = 0.819), width (r = 0.798), depth (r = 0.767), and the number of minifigures (r = 0.779). These results suggest that the physical size and contents of a LEGO set are closely associated with its retail price. The minimum recommended age also has a moderate positive correlation with price (r = 0.573), indicating that sets designed for older builders may tend to be more expensive. In contrast, user ratings (r = 0.252) and review counts (r = 0.260) have relatively weak correlations with price, suggesting that customer ratings and the number of reviews are less strongly associated with retail price than the physical characteristics of a set.

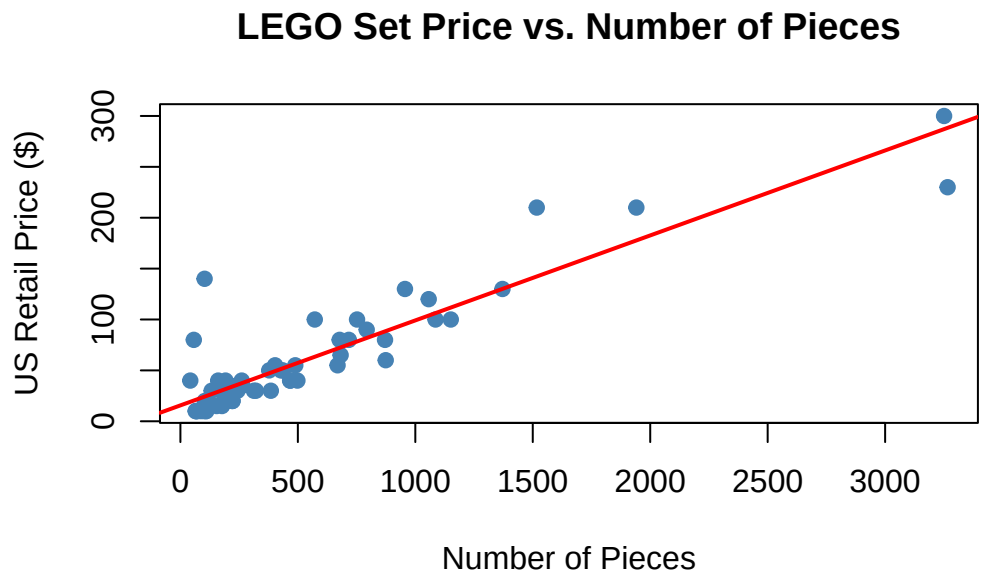
The matrix also reveals strong correlations among several predictor variables. For example, piece count and weight have a correlation of 0.866, while weight is also strongly correlated with height (r = 0.846), width (r = 0.827), and depth (r = 0.827). These relationships suggest that these variables could create multicollinearity in a multiple linear regression model. This means that including all these predictors simultaneously may make it harder to distinguish their individual contributions to price. Additionally, ratings and review counts have a moderate positive correlation (r = 0.594), suggesting that sets with more reviews may also tend to have higher ratings. Overall, the correlation matrix suggests that weight and piece count are promising predictors of retail price, although further regression analysis and model selection will be necessary to determine which variables provide the most useful independent explanatory power.

```r
#Scatter plots of key relationships

plot(
  US_retailPrice ~ pieces,
  data = data_complete,
  main = "LEGO Set Price vs. Number of Pieces",
  xlab = "Number of Pieces",
```

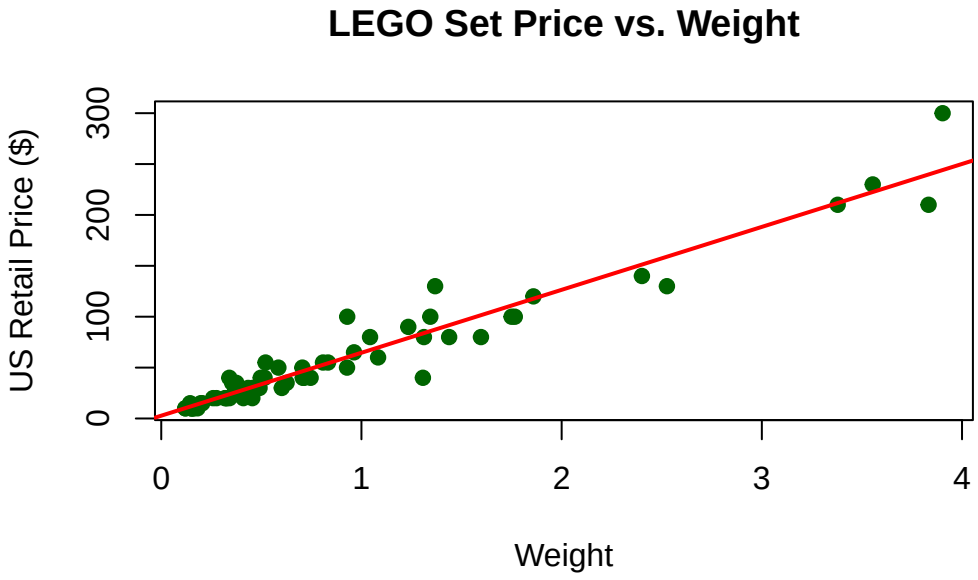
```
  ylab = "US Retail Price ($)",
  pch = 19,
  col = "steelblue"
)

# Add a fitted linear regression line
abline(
  lm(US_retailPrice ~ pieces, data = data_complete),
  col = "red",
  lwd = 2
)
```



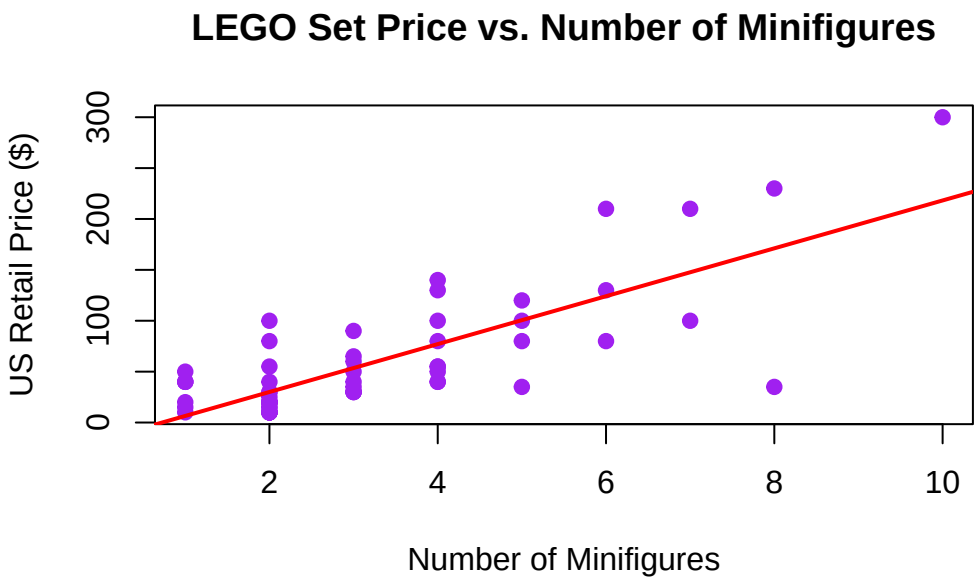

```
plot(
  US_retailPrice ~ weight,
  data = data_complete,
  main = "LEGO Set Price vs. Weight",
  xlab = "Weight",
  ylab = "US Retail Price ($)",
  pch = 19,
  col = "darkgreen"
)
```

```r
abline(
  lm(US_retailPrice ~ weight, data = data_complete),
  col = "red",
  lwd = 2
)
```




```r
plot(
  US_retailPrice ~ minifigs,
  data = data_complete,
  main = "LEGO Set Price vs. Number of Minifigures",
  xlab = "Number of Minifigures",
  ylab = "US Retail Price ($)",
  pch = 19,
  col = "purple"
)

abline(lm(US_retailPrice ~ minifigs, data = data_complete),col = "red",lwd = 2)
```



The scatterplots reveal positive relationships between LEGO set retail price and the number of pieces and weight. The strongest visual relationship appears in the plot of retail price versus weight, where price generally increases as the weight of a set increases. This is consistent with the correlation coefficient of 0.967, indicating a very strong positive linear association. The plot of retail price versus number of pieces also shows a clear upward trend, consistent with its correlation of 0.911. Sets with more pieces generally have higher retail prices, although there is some variation among sets with similar piece counts. Several observations lie farther from the regression line, suggesting that some sets are considerably more expensive or less expensive than their piece counts alone might predict. These observations may reflect differences in set design, theme, or other characteristics that influence pricing. The plots support the correlation analysis by indicating that physical characteristics, particularly weight and piece count, are strongly associated with LEGO set retail price. However, the presence of outlying observations and variation around the regression lines suggests that additional variables may help explain differences in price.

The relationship between retail price and the number of minifigures is also positive, but the pattern is less consistent than those observed for weight and piece count. Although sets with more minifigures tend to have higher prices, there is considerable variation in price among sets containing the same number of minifigures. For example, sets containing six minifigures have substantially different retail prices, suggesting that minifigure count alone does not fully explain a set's price. Overall, the scatterplots support the correlation analysis by indicating that physical characteristics, particularly weight and piece count, are strongly associated with LEGO set retail price. However, the presence of outlying observations and variation around the regression lines suggests that additional variables may help explain differences in price.

6. Best Subsets Regression

Best Subsets Regression is a model selecting method that takes the response variable, in this case US_retailPrice, and the predictors variables within the dataset and yields the possible models and their statistics. This method evaluates every possible combination of predictors and identifies the best model for each model size using adjusted R-squared and BIC. Using the data_complete where the missing value points are omitted.

```
 # load package for best subsets regression
library(leaps)
```

```
Warning: package 'leaps' was built under R version 4.5.3
```

```
# predictor variables
variable_names <- c("pieces", "minifigs", "weight", "height", "width", "depth",
                    "agerange_min", "rating", "reviewCount")

# run best subsets
best_subsets <- regsubsets(US_retailPrice ~ pieces + minifigs + weight +
                             height + width + depth + agerange_min + rating +
                             reviewCount, data = data_complete)


# display summary of the best regression models
best_summary_model <- summary(best_subsets)
best_summary_model
```

```
Subset selection object
Call: regsubsets.formula(US_retailPrice ~ pieces + minifigs + weight +
    height + width + depth + agerange_min + rating + reviewCount,
    data = data_complete)
9 Variables  (and intercept)
             Forced in Forced out
pieces           FALSE      FALSE
minifigs         FALSE      FALSE
weight           FALSE      FALSE
height           FALSE      FALSE
width            FALSE      FALSE
depth            FALSE      FALSE
agerange_min     FALSE      FALSE
rating           FALSE      FALSE
```

```
reviewCount      FALSE      FALSE
1 subsets of each size up to 8
Selection Algorithm: exhaustive
         pieces minifigs weight height width depth agerange_min rating
1  ( 1 ) " "    " "      "*"    " "    " "   " "   " "           " "
2  ( 1 ) " "    " "      "*"    " "    " "   " "   "*"           " "
3  ( 1 ) " "    "*"      "*"    " "    " "   " "   "*"           " "
4  ( 1 ) " "    "*"      "*"    " "    " "   " "   "*"           " "
5  ( 1 ) " "    "*"      "*"    " "    " "   "*"   "*"           "*"
6  ( 1 ) " "    "*"      "*"    " "    " "   "*"   "*"           "*"
7  ( 1 ) "*"    "*"      "*"    " "    " "   "*"   "*"           "*"
8  ( 1 ) "*"    "*"      "*"    "*"    " "   "*"   "*"           "*"
         reviewCount
1  ( 1 ) " "
2  ( 1 ) " "
3  ( 1 ) " "
4  ( 1 ) "*"
5  ( 1 ) " "
6  ( 1 ) "*"
7  ( 1 ) "*"
8  ( 1 ) "*"
```

```
# adjr2 tells us how well the predictors explain the variation in the
# response variable and the higher the adjr2 the better
best_summary_model$adjr2
```

```
[1] 0.9343154 0.9569526 0.9614188 0.9643101 0.9667275 0.9669445 0.9666173
[8] 0.9666360
```

```
#Cp tells how well a model balances fit and complexity
# the lower the Cp the better the model
best_summary_model$cp
```

```
[1] 56.953050 18.415108 11.663737  7.768334  4.809390  5.486479  7.024448
[8]  8.014201
```

```
#BIC tells us how simple and interpretable a model is
# the lower the BIC the better the model
best_summary_model$bic
```

```
[1] -158.8998 -181.6080 -185.2398 -186.9604 -188.2270 -185.6345 -182.0630
[8] -179.1483
```

```
#which tells which predictors are included in each model
best_summary_model$which
```

```
  (Intercept) pieces minifigs weight height width depth agerange_min rating
1        TRUE  FALSE    FALSE   TRUE  FALSE FALSE FALSE        FALSE  FALSE
2        TRUE  FALSE    FALSE   TRUE  FALSE FALSE FALSE         TRUE  FALSE
3        TRUE  FALSE     TRUE   TRUE  FALSE FALSE FALSE         TRUE  FALSE
4        TRUE  FALSE     TRUE   TRUE  FALSE FALSE FALSE         TRUE  FALSE
5        TRUE  FALSE     TRUE   TRUE  FALSE FALSE  TRUE         TRUE   TRUE
6        TRUE  FALSE     TRUE   TRUE  FALSE FALSE  TRUE         TRUE   TRUE
7        TRUE   TRUE     TRUE   TRUE  FALSE FALSE  TRUE         TRUE   TRUE
8        TRUE   TRUE     TRUE   TRUE   TRUE FALSE  TRUE         TRUE   TRUE
  reviewCount
1       FALSE
2       FALSE
3       FALSE
4        TRUE
5       FALSE
6        TRUE
7        TRUE
8        TRUE
```

Model Statistics

- Best adjusted R-squared = 0.9667 model 5
- Best Cp = lowest one 4.81 model 5
- Best BIC = -188.23 model 5

The optimal model is model 5 that includes minifigs, weight, depth, agerange_min and rating. This model has the highest adjusted R-squared of 0.9667, the lowest Cp of 4.91, and the lowest BIC of -188.23, showing that this model explains that variation in retail price of LEGO sets the best and with relatively low complexity. Variables such as pieces, height, width, and reviewCount were not selected due to strong correlation with weight.